

The Effective Viscosity for a Suspension of Spheres

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A formula is proposed for the effective relative shear viscosity η_r of a suspension of solid spheres. This formula is a modification of a formula first given by Saitô to which a function of the volume fraction is added containing the effects due to correlations and the hydrodynamic interactions between the spheres. The theoretical background of this formula in which the effective self-correlations are taken into account to all orders was given in an earlier paper. A simple derivation is given for the precise manner in which Saitô's formula is modified which brings out the analogy with Clausius and Mossotti's formula for the dielectric constant more clearly. Comparison with recent experimental data for uncharged spheres shows that an expansion of $(\eta_r - 1)/(\eta_r + \frac{5}{2})$ to quadratic order in the volume fraction is sufficient to fit the experimental data in the high shear limit for all volume fractions observed. One may calculate this coefficient from the value for the quadratic coefficient in an expansion of the effective viscosity given by Batchelor and Green. Within the accuracy of the theoretical value this coefficient is in reasonable agreement with the experimental value. In the low shear limit an expansion to third order in the volume fraction is needed to obtain a comparable fit. All theoretical results for the viscosity in this limit are at best in qualitative agreement with the experimental data. © 1987 Academic Press, Inc.

1. INTRODUCTION

In a recent paper de Kruif *et al.* (1) present experimental results for the viscosity of a dispersion of hard silica spheres as a function of both the shear rate and the volume fraction ϕ for $0.0006 \leq \phi \leq 0.6$. Their results agree with the experimental results found by Krieger (2) on a highly cross-linked latex dispersed at high ionic strengths which also closely resembles a hard sphere system. The experimental values of the relative viscosity are found to be described up to $\phi \approx 0.35$ by

$$\eta_0 = 1 + \frac{5}{2}\phi + (4 \pm 2)\phi^2 + (42 \pm 10)\phi^3 \quad [\text{A}]$$

in the low shear limit and by

$$\eta_\infty = 1 + \frac{5}{2}\phi + (4 \pm 2)\phi^2 + (25 \pm 7)\phi^3 \quad [\text{B}]$$

in the high shear limit. While the value of the coefficient of the quadratic term agrees with the theoretical predictions (3–6) it is not sufficiently accurate to decide which theory fits the experimental data best. The large size of

the third-order terms in the above equations as well as the 50% error in the coefficient of the second-order term indicates that the convergence of an expansion of the viscosity in the volume fraction is slow. The use of such an expansion is certainly inappropriate for $\phi > 0.35$, where the viscosity appears to diverge at $\phi_{s,0} = 0.63 \pm 0.02$ in the low shear limit and at $\phi_{s,\infty} = 0.71 \pm 0.02$ in the high shear limit. Various theories have been presented to explain the spectacular increase of the viscosity as a function of the volume fraction. The simplest such theory is a typical effective medium theory (7, 8) and leads to a formula first given by Lundgren (9) on the basis of very different arguments. This theory gives a divergence at $\phi_s = 0.4$ and, in general, grossly overestimates the value of the viscosity for $\phi > 0.25$. Beenakker (10) uses a formal solution which accounts only for hydrodynamic interactions between an arbitrary number of spheres to derive an expansion for the viscosity in fluctuations of the volume fraction around its av-

erage value. The viscosity is then calculated numerically to second order in this expansion using Percus-Yevick correlation functions for hard spheres. In a completely different approach Russel and Gast (19) neglect many particle hydrodynamic interactions but evaluate the various contributions to the viscosity given already by Batchelor *et al.* (4, 6) using volume fraction-dependent correlation functions at higher volume fractions. In particular they show that the thermodynamic stresses, which are important in the low shear limit, increase dramatically as a function of the volume fraction. As we will discuss the agreement of neither of the theories for higher volume fractions is particularly satisfactory.

In this paper we propose the use of the following formula for the relative viscosity

$$\frac{\eta_r - 1}{\eta_r + 3/2} = \phi[1 + S(\phi)]. \quad [1.1]$$

If one sets $S = 0$ one obtains an expression given originally by Saitô (11, 12). Saitô's formula is analogous to the Clausius-Mossotti equation for the permittivity of a dielectric (13); in both formulas contributions due to interactions and correlations between two or more spheres are neglected. The term proportional to $S(\phi)$ on the right-hand side of the above formula accounts for these contributions. A theoretical derivation of this formula is given by Bedeaux, *et al.* (5, 14) who give an expansion for $(\eta_r - 1)/(\eta_r + \frac{3}{2})$ and thus for $S(\phi)$ in powers of ϕ (15). In this derivation contributions of the thermodynamic stresses due to diffusion (6) are neglected and therefore their derivation is adequate for only a sufficiently high Peclet number. They do take hydrodynamic interactions between arbitrary numbers of spheres into account. As is discussed, for instance, by Hinch (20) these many-particle hydrodynamic interactions lead to an expansion of the effective viscosity in the volume fraction if one averages the conservation laws over an ensemble of realizations. Using a similar procedure together with

the method of induced forces Bedeaux *et al.* (5) show how self-correlations may be accounted for to an arbitrary order. This then leads to a resummation of the expansion of the effective viscosity which is most conveniently given using Eq. [1.1] and by giving the expansion for $S(\phi)$. It should be stressed that both the expansion of $S(\phi)$, as well as the expansion of the effective viscosity, are systematic expansions and would give the same value of the effective viscosity if the expansion coefficients are known to an arbitrary order. In practice, however, one is only able to calculate either the effective relative viscosity η_r or $(\eta_r - 1)/(\eta_r + \frac{3}{2})$ to second order in ϕ . The coefficients are, of course, trivially related. The claim to be analyzed in this paper is the fact that due to taking self-correlations into account to all orders the expansion of $(\eta_r - 1)/(\eta_r + \frac{3}{2})$ converges much better. In fact we find that using the quadratic coefficient which follows from Batchelor and Green (4) in Eq. [1.1] one finds values of η_r which agree with the experimental values (1, 2) within their accuracy for almost all volume fractions. There is, of course, a large literature on the behavior of the effective viscosity of suspensions (see, e.g., references given in Refs. (5, 10, 19, 20)). As nobody addresses the point made in this paper and since the agreement of these theories is at best qualitative for volume fractions which are not small, we will not discuss this literature in detail. We will use Eq. [1.1], together with an expansion of $S(\phi)$ also, to analyze the experimental results for low Peclet numbers. It should be emphasized that in that case the use of this equation is not based on a solid theoretical analysis as it is for high Peclet numbers.

Bedeaux *et al.* (5) find using a multipole expansion of the induced force density to third order

$$S(\phi) = 0.92\phi + O(\phi^2). \quad [1.2]$$

In Fig. 1 we compare the various theoretical values with the low shear and the high shear experimental values. It is clear that Eq. [1.1], together with $S = 0.92\phi$, gives values with by

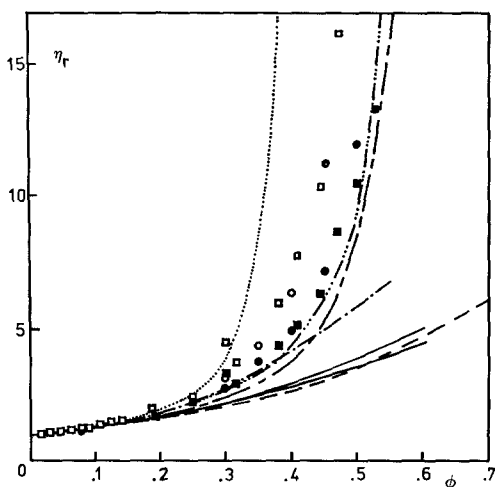


FIG. 1. Relative viscosity against volume fraction. De Kruif's *et al.* (1) (■) and Krieger's (2) (●) high shear limiting viscosity; de Kruif's *et al.* (1) (□) and Krieger's (2) (○) low shear limiting viscosity. Theoretical predictions: Saitô (11) (---); Lundgren (9) (···); Batchelor *et al.* (4, 6) solid lines with as coefficients 5.2 and 6.2; Beenakker (10) (---); Russel *et al.* (19) (---); Bedeaux *et al.* (5) (---).

far the best agreement for the high shear value for which the purely hydrodynamic value is most appropriate. None of the theories fit the experimental values for the low shear case in a satisfactory manner. Russel and Gast (19) show in a convincing way that thermodynamic stresses become very important in the low shear limit. Their analysis is, however, not systematic like the one given by Batchelor (6) who calculates this contribution to second order in ϕ . For higher volume fractions one should not only calculate how the pair correlation function is modified due to the shear to which they restrict themselves on the basis of the assumption that the hydrodynamic interaction is pairwise additive. Three and more particle hydrodynamic interactions clearly become important (10).

In the following section we shall further discuss the use of Eq. [1.1] for the analysis of the experimental results. We find that a good fit of the data is obtained using an expansion of $S(\phi)$ in ϕ for all values of ϕ . The results are

again compared with the various theoretical predictions. In the last section we give a simple derivation of Saitô's formula. An analysis of the dielectric case along the same lines gives the Clausius-Mossotti equation. The close analogy of the two derivations shows how contributions due to correlations in the distribution of the spheres will lead to similar corrections. The use of Eq. [1.1] for the viscosity then seems the natural way to account for these additional contributions as is done in a similar way for the dielectric constant.

2. ANALYSIS OF EXPERIMENTAL AND THEORETICAL RESULTS USING $S(\phi)$

In Fig. 2 the experimental (1) and some of the theoretical values of the function $S(\phi)$ have been plotted. For small values of ϕ the experimental accuracy of $\eta_r - 1$ and consequently of $S(\phi)$ as calculated using Eq. [1.1] is unsatisfactory which explains the large scatter in

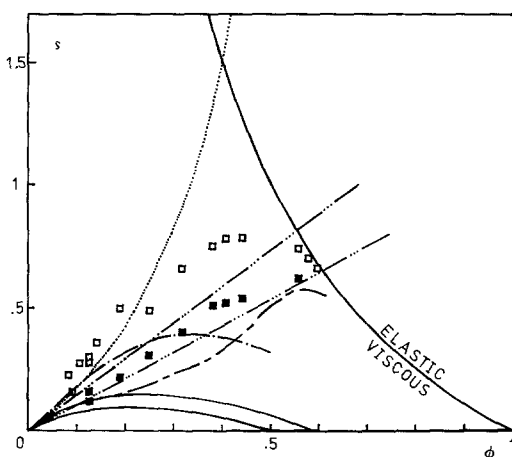


FIG. 2. S as function of the volume fraction. High shear limiting values (■) and low shear limiting values (□) which follow from the data from de Kruif *et al.* (1). Theoretical predictions: Lundgren (9) (···); Batchelor *et al.* (4, 6) if one uses the expansion of η_r to second order in the volume fraction solid lines; expansion of S to linear order in ϕ with coefficient calculated from the 5.2 and 6.2 given by Batchelor *et al.* (4, 6) (---); Beenakker (10) (---); Russel *et al.* (19) (---). The boundary between the viscous and the elastic region is also indicated by a solid line.

the experimental points for $\phi < 0.1$. For larger values of ϕ the experimental values can be fitted using an expansion of S in the volume fraction. For the high shear value of the viscosity we find that a satisfactory fit of S is obtained using a straight line

$$S(\phi) = (1.21 \pm 0.03)\phi. \quad [2.1]$$

In Fig. 3 we plotted the value of the relative viscosity found upon substitution of Eq. [2.1] into Eq. [1.1] as well as the experimental values. A good fit of the data is thus obtained for all volume fractions if one uses Eq. [1.1] with only one adjustable parameter (the slope) in the high shear limit. The theoretical values for this slope is 0.92 if one uses a multipole expansion of the force density to 3rd order (5) and improves to 1.14 if one uses a multipole expansion to 15th order (14). If one calculates the slope using the second-order coefficient 5.2 ± 0.3 for an expansion of μ_r in ϕ given by Batchelor and Green (4) one finds 1.08 ± 0.12 which is the value one would also find if one

goes to arbitrary order in the multipole expansion.

We conclude that within the same accuracy the experimental and the theoretical values for the slope are in reasonable agreement with each other. The fact that $S(\phi)$ can be fitted using a straight line is surprising. Brady's (21) simulations show that the radial distribution function at high Peclet numbers is not uniform while, furthermore, many particle hydrodynamic interactions become important. The modification of the radial distribution function will lead to a decrease of η_r (shear thinning) while the contributions to η_r due to many particle hydrodynamic interactions will increase η_r (10). Why the combination of these effects gives such a small coefficient of the quadratic term in the expansion of $S(\phi)$ for high Peclet numbers but not for low Peclet numbers (see below) is unclear.

For the low shear value of the viscosity we find a satisfactory fit of S using a parabola:

$$S(\phi) = (3.08 \pm 0.11)\phi - (3.15 \pm 0.21)\phi^2. \quad [2.2]$$

In Fig. 3 we plotted the value of the relative viscosity found upon substitution of Eq. [2.2] into Eq. [1.1] as well as the experimental values. A good fit of the data for all volume fractions is thus obtained for the low shear limit using two adjustable parameters. The linear coefficient may be calculated using the second-order coefficient 6.2 for an expansion of μ_r in ϕ given by Batchelor (6) which gives 1.48. In this case the theoretical results are at best in qualitative agreement with the experimental results.

If the relative viscosity approaches infinity at a volume fraction ϕ_s one may use Eq. [1.1] to derive the following relation between the value of $S_s \equiv S(\phi_s)$ and ϕ_s

$$S_s = \frac{1}{\phi_s} - 1. \quad [2.3]$$

We have plotted this line in Fig. 2 where it serves as a rough boundary between the lower values of ϕ for which the suspension behaves as a viscous medium and the higher values of

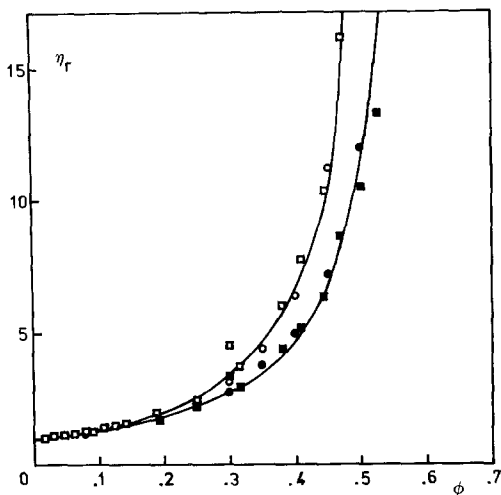


FIG. 3. Relative viscosity against volume fraction. De Kruif's *et al.* (1) (■) and Krieger's (2) (●) high shear limiting viscosity; de Kruif's *e al.* (1) (□) and Krieger's (2) (○) low shear limiting viscosity. The solid lines give the fit obtained using the proposed formula (1.1) together with Eqs. (2.1) and (2.2) for S .

ϕ where the suspension behaves as an elastic medium. If the relative viscosity does not diverge S must satisfy

$$S(\phi) < \frac{1}{\phi} - 1 \quad \text{for all } \phi. \quad [2.4]$$

As Eqs. [2.1] and [2.2] overestimate the value of $S(\phi)$ close to ϕ_s , substitution of these equations in Eq. [2.3] gives values of ϕ_s which are somewhat lower than the experimental values.

3. ON THE DERIVATION OF THE GENERALIZATION OF SAITÔ'S FORMULA

In order to derive a generalization of Saitô's formula we first consider a spherical droplet containing a fluid with viscosity η_1 dispersed in a fluid with viscosity η_0 . Both fluids are incompressible. When the velocity field is zero there is a small pressure difference in the two fluids which is related to the surface tension by Laplace's law. We are interested in the response of the system to an "incident" elongational flow field $\vec{\alpha} \cdot \mathbf{r}$ where $\vec{\alpha}$ is a symmetric traceless tensor. The origin of the coordinate frame has been chosen in the center of the sphere. At the surface of the sphere both the velocity field $\mathbf{v}$ and the force on the surface of the sphere $\vec{\mathbf{P}} \cdot \mathbf{n}$ should be continuous. $\vec{\mathbf{P}}$ is the sum of the viscous pressure tensor and the change of the hydrostatic pressure from the value for $\vec{\alpha} = 0$ times the unit tensor. As the problem is solved in the linear regime the boundary conditions may be imposed at the spherical surface $|\mathbf{r}| = R$ of the droplet. The general form of the velocity field outside the sphere is of the form (16)

$$\mathbf{v}(\mathbf{r}) = \vec{\alpha} \cdot \mathbf{r} - \frac{5}{2} A \frac{R^3}{r^2} \vec{\alpha} : \hat{r} \hat{r} \hat{r} + B \frac{R^5}{r^4} \vec{\alpha} : \hat{r} \left(\frac{5}{2} \hat{r} \hat{r} - \hat{\mathbf{I}} \right) \quad [3.1]$$

while the change of the hydrostatic pressure field outside the sphere is given by

$$p(\mathbf{r}) = -5\eta_0 A \left(\frac{R}{r} \right)^3 \vec{\alpha} : \hat{r} \hat{r}, \quad [3.2]$$

where $r \equiv |\mathbf{r}|$ and $\hat{r} \equiv \mathbf{r}/r$. Inside the sphere the velocity field is given by

$$\mathbf{v}(\mathbf{r}) = C \vec{\alpha} \cdot \mathbf{r} \quad [3.3]$$

while the change of the hydrostatic pressure field is zero. The constants A , B , and C follow from the boundary conditions at the surface of the sphere. One finds

$$A = B = 1 - C = \frac{\eta_1 - \eta_0}{\eta_1 + (3/2)\eta_0}. \quad [3.4]$$

If the sphere is solid one may find the various fields using $\eta_1 = \infty$. This gives $A = B = 1$ and $C = 0$ which reduces the fields given in Eqs. [3.1] and [3.2] to the fields given, e.g., in Ref. (16).

If one uses Eq. [3.1] to calculate the effective viscosity for a low volume fraction ϕ of suspended spheres one obtains in the usual way (16)

$$\eta_{\text{eff}} = \eta_0 \left(1 + \frac{5}{2} A \phi \right) = \eta_0 \left(1 + \frac{5}{2} \frac{\eta_1 - \eta_0}{\eta_1 + (3/2)\eta_0} \phi \right). \quad [3.5]$$

In the limit $\eta_1 \rightarrow \infty$ this reduces to the expression given by Einstein (17).

In order to obtain a formula for larger volume fractions we place the spherical droplet with a viscosity η_1 in a concentric shell of fluid with a viscosity η_0 . The outer radius R_0 of the concentric shell is chosen such that the volume fraction is given by $\phi = (R/R_0)^3$. Thus the relative amount of the two fluids inside the outer shell is the same as in the fluid as a whole. This system of two concentric shells is placed in a hypothetical fluid with the effective viscosity of the suspension. This effective viscosity is then chosen such that the incident velocity field $\vec{\alpha} \cdot \mathbf{r}$ induces no net force moment in the concentric shells. We will make this definition more precise below. For the time being we will take the viscosity outside the outer shell equal to η and calculate the velocity field and the contribution to the hydrostatic pressure proportional to $\vec{\alpha}$. The general form of the velocity field is

$$\mathbf{v}(\mathbf{r}) = \begin{cases} \begin{aligned} &\vec{\alpha} \cdot \mathbf{r} - \frac{5}{2} A \frac{R_0^3}{r^2} \vec{\alpha} : \hat{r} \hat{r} \hat{r} \\ &+ B \frac{R_0^5}{r^4} \vec{\alpha} : \hat{r} \left(\frac{5}{2} \hat{r} \hat{r} - \mathbf{I} \right) \end{aligned} & \text{for } R_0 < r \\ \begin{aligned} &C_0 \vec{\alpha} \cdot \mathbf{r} - \frac{5}{2} A_0 \frac{R^3}{r^2} \vec{\alpha} : \hat{r} \hat{r} \hat{r} \\ &+ B_0 \frac{R^5}{r^4} \vec{\alpha} : \hat{r} \left(\frac{5}{2} \hat{r} \hat{r} - \mathbf{I} \right) \end{aligned} & \text{for } R < r < R_0 \\ C_1 \vec{\alpha} \cdot \mathbf{r} & \text{for } r < R \end{cases} \quad [3.6]$$

and the general form of the hydrostatic pressure field is

$$p(\mathbf{r}) = \begin{cases} -5\eta A \left(\frac{R_0}{r} \right)^3 \vec{\alpha} : \hat{r} \hat{r} & \text{for } R_0 < r \\ -5\eta_0 A_0 \left(\frac{R}{r} \right)^3 \vec{\alpha} : \hat{r} \hat{r} & \text{for } R < r < R_0 \\ 0 & \text{for } r < R. \end{cases} \quad [3.7]$$

Using the boundary conditions in $r = R$ one obtains analogous to Eq. [3.4]

$$A_0 = B_0 = C_0 - C_1 = C_0 \frac{\eta_1 - \eta_0}{\eta_1 + (3/2)\eta_0}. \quad [3.8]$$

The boundary conditions in $r = R_0$ subsequently give

$$A = \frac{(\eta_0 - \eta)(\eta_1 + (3/2)\eta_0) + \phi(\eta_1 - \eta_0)(\eta + (3/2)\eta_0)}{(\eta_0 + (3/2)\eta)(\eta_1 + (3/2)\eta_0) + (3/2)\phi(\eta_0 - \eta)(\eta_1 - \eta_0)} \quad [3.9]$$

$$B = 1 - C_0 \left(1 - \phi^{5/3} \left(\frac{\eta_1 - \eta_0}{\eta_1 + (3/2)\eta_0} \right) \right) \quad [3.10]$$

$$C_0 = \frac{5}{2} \frac{\eta(\eta_1 + (3/2)\eta_0)}{(\eta_0 + (3/2)\eta)(\eta_1 + (3/2)\eta_0) + (3/2)\phi(\eta_0 - \eta)(\eta_1 - \eta_0)}. \quad [3.11]$$

The force moment of the two concentric spheres which contributes to the average viscous pressure is zero if $A = 0$. We now choose $\eta = \eta_{\text{eff}}$ such that this is the case. This gives

$$\frac{\eta_{\text{eff}} - \eta_0}{\eta_{\text{eff}} + (3/2)\eta_0} = \phi \frac{\eta_1 - \eta_0}{\eta_1 + (3/2)\eta_0}. \quad [3.12]$$

In the limit $\eta_1 \rightarrow \infty$ one obtains

$$\frac{\eta_{\text{eff}} - \eta_0}{\eta_{\text{eff}} + (3/2)\eta_0} = \phi \quad \eta_{\text{eff}} = \eta_0 \frac{1 + (3/2)\phi}{1 - \phi} \quad [3.13]$$

which is a formula first given by Saitô (11, 12). Equation [3.12] generalizes this formula to finite values of η_1 .

It is clear from the above derivation that details about the distribution of the dispersed spheres with viscosity η_1 in a fluid with viscosity η_0 have been neglected. Saitô's formula corresponds to a model in which the velocity field far away from the dispersed spheres propagates through the average medium with viscosity η_{eff} while closer to the sphere these fields enter a concentric region with a viscosity η_0 . They are accordingly modified before they reach the suspended sphere.

This situation is similar to the calculation of the effective dielectric constant of a suspension of dielectric or conducting spheres in some other dielectric material. An analysis along the same lines as above leads to the well-known Clausius-Mossotti formula (18). As in the dielectric case one could reformulate the above derivation of Saitô's formula using the concept of a local field (12, 8). The response of the spheres should then be calculated in this local field rather than in the incident field.

Equation [1.1] in which $\eta_r \equiv \eta_{\text{eff}}/\eta_0$ is a direct generalization of Saitô in which the contributions due to correlations are added in a way which preserves the crucial difference between the local and the incident field. In the dielectric case this is done in the same way and for exactly the same reason. While the

discussion in this section tries to clarify why we propose the use of Eq. [1.1] for the analysis of the experimental results it does not give explicit expressions for $S(\phi)$ in terms of correlation functions. Such expressions were already given, however, in a much more formal context in Ref. (5) for the case where only hydrodynamic interactions (to all orders) are taken into account (see also [15]). In conclusion we feel that theoretical arguments together, in particular, with the usefulness in the interpretation of the experimental results discussed in the previous section strongly support the use of Eq. [1.1] together with expansions of $S(\phi)$ in the volume fraction. The high shear case where the theoretical value for the coefficient of the linear term in this expansion gives viscosities which are in remarkable agreement with the experimental data further supports this point of view.

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